

Elbow Flab Stretcher: An Auditory-Induced Skin Stretch Illusion via Audio-Tactile Synchrony

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Abstract

Recent research demonstrates that body image and ownership can be flexibly modulated through multisensory integration, exemplified by the Rubber Hand Illusion (RHI). Unlike proprioception, skin lacks precise spatial sensors, leading to ambiguous spatial perception. Building on this, the Slime Hand Illusion (SHI) induces a sensation of skin stretching via visuotactile stimulation. This study introduces a novel illusion, the Elbow Flab Stretcher (referred to in Japanese as “*Otohifu*”), which replaces visual cues with auditory frequency modulation synchronized with tactile skin stretching on the elbow. Twenty participants underwent conditions manipulating visual input and sound frequency modulation. Results showed that auditory frequency changes corresponding to the physical elevation changes of the sound source significantly enhanced the sensation of skin elongation, especially with eyes closed. These findings suggest that auditory cues alone can modulate body perception, highlighting audio-tactile integration as a contributor to body representation. The study broadens understanding of multisensory effects on body image and encourages further research into neural mechanisms and applications across body parts and sensory modalities.

CCS Concepts

• Human-centered computing → Sound-based input / output.

Keywords

Rubber Hand Illusion; Slime Hand Illusion; Skin Stretching; Audio-Tactile Integration

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1 Introduction

Recent studies, including the well-known Rubber Hand Illusion (RHI [1]), have demonstrated that body image and the sense of bodily ownership—the feeling that “this is my body”—can be flexibly modulated through the integration of multiple sensory modalities. However, the induction of RHI typically requires certain spatial and morphological constraints, such as a proximity of approximately 20 cm between the rubber hand and the real hand, as well as visual similarity between the two [4, 8].

In contrast, unlike proprioception, the skin lacks spatial position sensors, and thus the perception of where the skin is located in space tends to be ambiguous. Focusing on this characteristic, our laboratory proposed the Slime Hand Illusion (SHI [3]), in which participants feel as though their skin is being stretched far beyond its physical limits. SHI is induced via visuotactile stimulation using a mirror visual feedback [5] setup, and participants have been reported to experience skin deformation of up to 30 cm.

Moreover, SHI can be elicited even without using actual slime. Merely mimicking the pinching-and-pulling motion in empty space can induce the sensation that the skin is being stretched—an effect referred to as the Invisible Slime Hand Illusion. A related variation that applies this non-visual version to the earlobe is known as the Buddha’s Ear Illusion [2]. In this illusion, participants feel as if their earlobe is dramatically stretched—often over 20 cm—despite only slight physical pulling. This sensation is induced through the synchronization of tactile stimulation with visual gestures mimicking the pulling motion. These results demonstrate that the illusory skin deformation observed in SHI can also occur in the earlobe, supporting the notion that visuotactile synchrony can flexibly alter body image even in skin regions.

Taken together, these findings suggest that SHI, unlike RHI, can occur beyond the conventional constraints of spatial proximity and morphological similarity, allowing for more flexible modulations of body ownership.

As described above, both SHI and RHI involve transformations of body image through visuotactile synchrony. In contrast, recent research has shown that synchronous auditory and tactile stimulation can also modulate body image, for example by inducing the sensation of an elongated arm [6, 7].

However, the effects of auditory–tactile integration on body image—particularly in relation to the skin—remain largely unexplored. In this study, we propose a novel illusion, *Otohifu*, which adapts the framework of SHI by replacing the visual stimuli with auditory cues (Figure 1). We aim to investigate whether this auditory-based illusion can similarly induce a perceived deformation of the skin.

2 Method

Twenty university students participated in the experiment.

The sound source device used was "Otamamin" (Meiwa Denki), equipped with a light sensor that modulates sound frequency according to light intensity. For example, the closer the device is to a light source, the higher the emitted sound frequency; conversely, moving away results in a lower frequency. The light stimulus was provided using a typical smartphone flashlight placed slightly above the floor and directed upward. The experimenter pinched and pulled the participant's right elbow skin with their left hand. Simultaneously, holding the sound source device in the right hand, the experimenter moved it closer to or farther from the light source to synchronize the tactile manipulation with auditory stimulation.

The experimental procedure began with 30 to 60 seconds of *Otohifu* stimulation, after which participants rated the sensation of their elbow skin stretching on a 7-point scale ranging from -3 (not at all) to +3 (very strong). Next, *Otohifu* stimulation was applied in three back-and-forth strokes between the elbow and the floor.



Figure 1: Subjective Body Deformation Induced by “*Otohifu*”

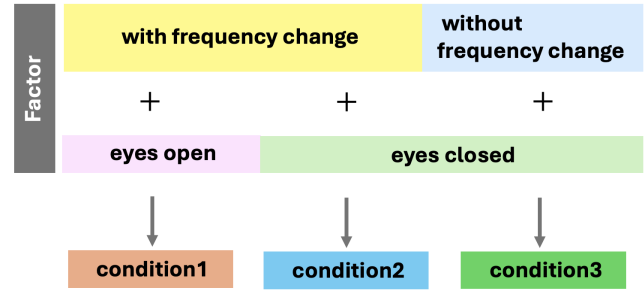


Figure 2: Experimental Condition

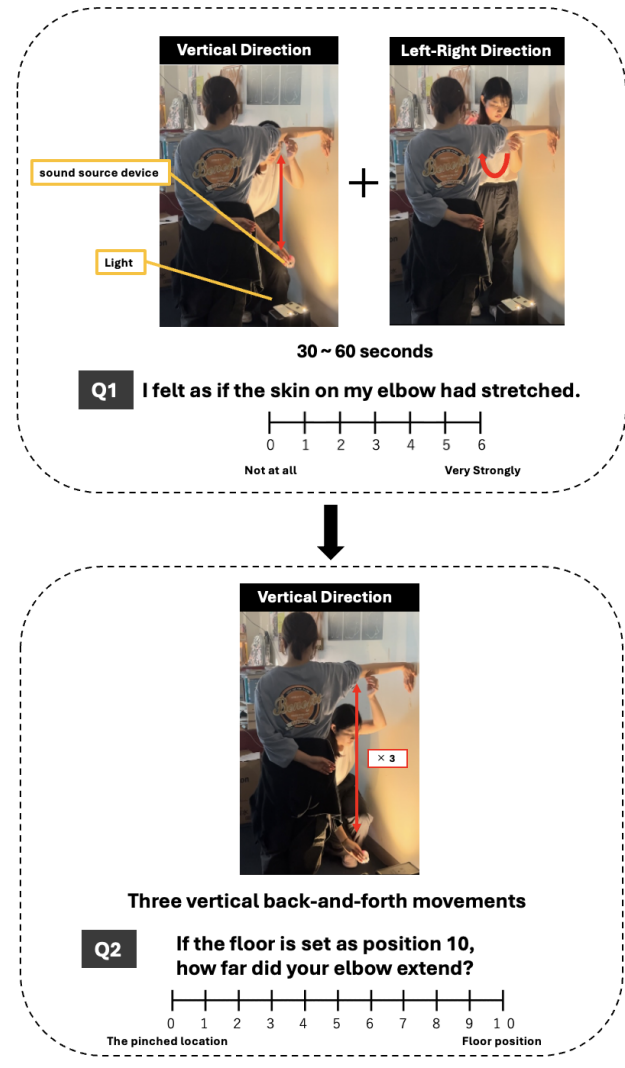


Figure 3: Experimental Procedure

Participants then rated the perceived elbow extension on an 11-point scale from 0 to 10, where the floor position was defined as 10 (Figure 3).

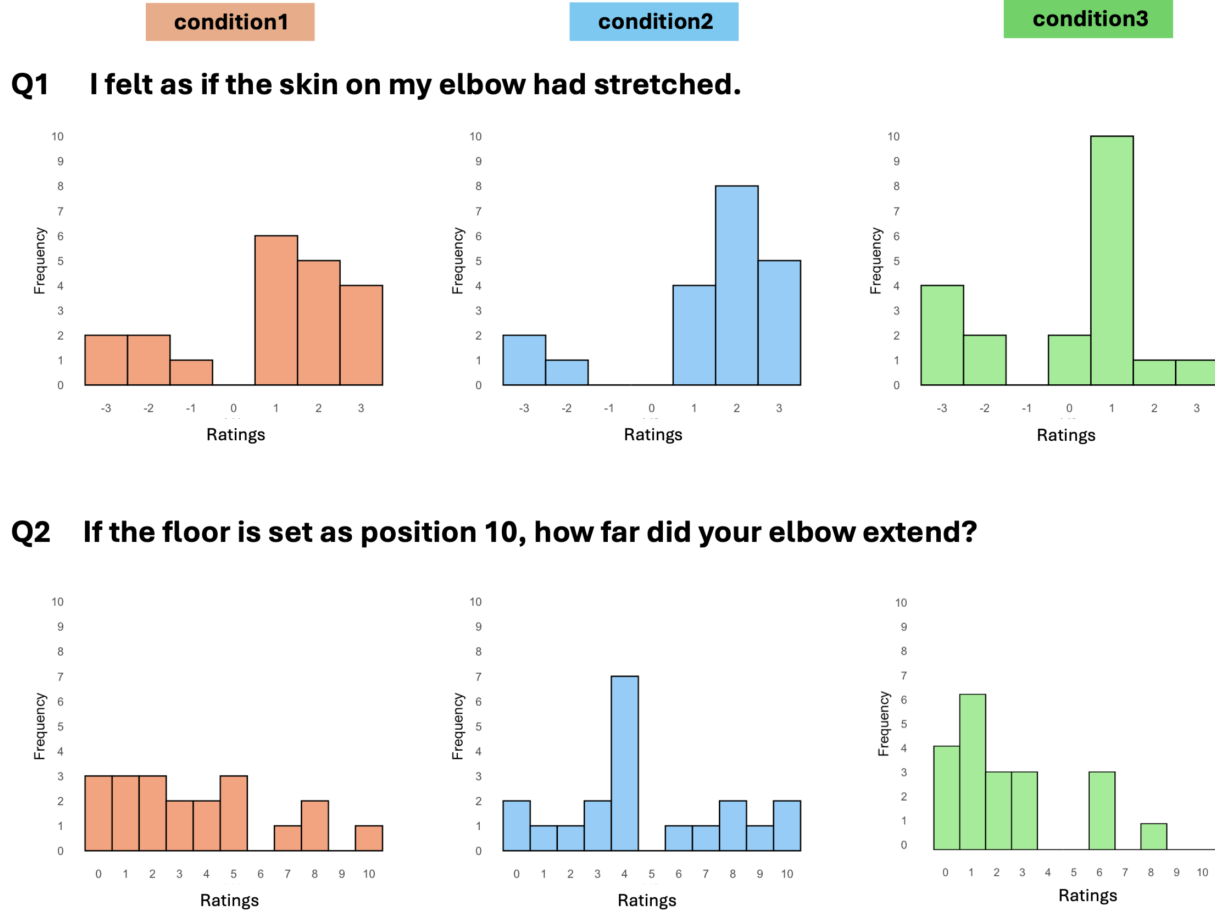


Figure 4: Histograms of Responses to the Two Questionnaire

3 Results

Histograms summarizing subjective scores from Q1 and Q2 are presented in Figure 4. Because the normality assumption was violated for these data, a non-parametric one-way ANOVA using the Friedman test was conducted. Post-hoc pairwise comparisons were performed using Wilcoxon signed-rank tests with Holm correction for multiple comparisons.

Significant differences were found between the eyes-closed/frequency-modulation and eyes-closed/no-frequency-modulation conditions for both Q1 ($\chi^2(2) = 17.45, p < .001$) and Q2 ($\chi^2(2) = 12.54, p < .01$). Specifically, under eyes-closed conditions, the frequency modulation condition elicited significantly higher subjective scores than the no-frequency-modulation condition (all $p < .01$).

Mean scores for Q1 were as follows: eyes open with frequency change: 0.85; eyes closed with frequency change: 1.35; eyes closed without frequency change: -0.05. For Q2, mean scores were eyes open with frequency change: 3.55; eyes closed with frequency change: 4.75; eyes closed without frequency change: 2.35.

4 Discussion

The present study aimed to investigate the subjective illusory effects of *Otohiifu* and to examine the fundamental conditions under which the illusion is induced, focusing on auditory frequency modulation and visual conditions.

The results demonstrated that the modulation of the stimulus sound frequency, that is, changes in pitch conveying some form of spatial or positional information, is crucial for eliciting the illusion. Furthermore, the sensation of skin stretching was rated highest under the eyes-closed condition.

These findings suggest that the sensation of skin elongation can arise solely from auditory information without relying on visual input. In other words, auditory spatial cues may contribute to the multisensory integration underlying body representation and modulate body image.

This study contributes to a deeper understanding of body perception alterations induced by audio-tactile integration through the novel *Otohiifu* illusion. Future research should further explore the underlying neural mechanisms, extend investigations to other body parts, and examine interactions with visual information.

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